University Of Oslo

FYS2150

Solar Cell

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Abstract

This project examines the behavior of solar cells under both uniform and non-uniform illumination conditions. The goal is to understand how solar cells can be practically used as sources of electrical power.

We first identify a single solar cell by measuring its current–voltage (I–V) curve in both dark and illuminated conditions . We then study how the power output varies with different resistances and when the cell produces the maximum power. We found that the power increases as we increase the load up to a certain point, after which it begins to decrease.

Next, two identical cells are connected in series and in parallel, and we compare their efficiency. In the final part, one of the cells is partially shaded to simulate conditions in reality where solar cells can be exposed to uneven light. We found that parallel-connected solar cells are more efficient than series-connected ones under different conditions.

1 Introduction

Extracting energy is one of the greatest challenges of our time. The demand for a low-cost and efficient energy source that does not harm the planet is currently a major challenge for our race .

Solar energy is one of the most accessible and sustainable energy sources available to us today. The sun is the main source of external energy that we receive on our planet, and it powers a large number of the planet's natural phenomena that can be used for our benefit . This makes the conversion of sunlight into electrical energy an interesting subject,

A solar cell is a semiconductor device that converts light into electrical energy based on the photoelectric effect. When photons with enough energy shines on the surface of the cell, they excite electrons causing them to be ejected from their atoms , running a current in an circuit. The efficiency of this process depends on several factors, including the wavelength of the light, the intensity of illumination, and the electrical load connected to the cell as well as how cells are connected together.

In this series of experiments, we will focus on the effect of resistance variation on the power delivered by a single cell. We will also study two types of connecting solar cells : series connecting and parallel connecting and compare their efficiency in both uniform and non-uniform light.

2 theory

The solar cells are made of a semiconductor material . A semiconductor is a material with electrical conductivity between that of a conductor and an insulator commonly based on silicon. . Silicon atoms have four valence electrons and form a crystal by sharing electrons with neighboring atoms, satisfying the 8-rule and creating strong bonds. When an electron gains enough energy, it can escape the bond and become a free electron, leaving behind a vacancy known as a "hole" needs to be filled .An electron from a neighboring atom will jump to fill this hole, creating a new hole in its original location . This causes both electrons and holes to move through the crystal, with holes moving in the opposite direction to the electrons. The more electrons that are released and holes that are created, the better is the conductivity.

To further improve conductivity, a process known as "doping "is used. This process is based on replacing approximately one in 10^6 silicon atoms with atoms of a different element.That allows us to improve conductivity without affecting the structure of the crystal.

Doping with atoms that have five valence electrons gives extra free electrons and creates an n-type semiconductor .Doping with atoms that have three valence electrons increases the number of holes and creates a p-type semiconductor. In solar cells a combination of these two types of semiconductor is used and it is called "pn-junction " as shown in the figure 1 below:

In the region between the p-side and n-side (called the transition area), there exists an electric field directed from the n-side to the p-side. This field

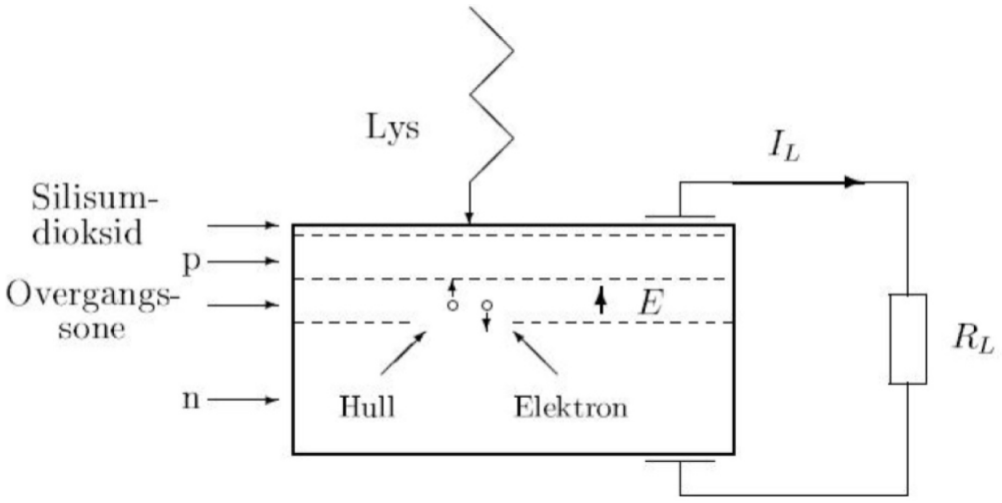


Figure 1: Solar cell as semiconductor

prevents further movement of charges from the n-region to the p-region.

In the absence of light, a solar cell behaves like a standard p–n junction. However, the situation changes when light with enough energy shines on the junction. Light striking the transition region will release bound electrons, creating free electrons and hence holes .The internal electric field then drives the electrons toward the n-region and the holes toward the p-region.

As a result, electrons gather in the n-region and holes in the p-region, a potential difference (voltage) is created across the junction, with the p-side becoming positively charged and the n-side negatively charged. Over time, the field in the transition zone decreases, and the flow of charge stops. To keep a continuous flow, a conductor is externally connected between the p- and n-regions. This allows electrons to continue flowing from the n-side to the p-side through the wire, producing an current in the circuit . The current will remain as long as light continues to reach the solar cell.

In our discussion, We will only be using formulas related to voltage, cur-

rent, and resistance. Therefore, **Ohm's law** is useful to our analysis :

$$I = \frac{V}{R} \tag{1}$$

Where :

- V: Is the voltage .
- R: is the resistance .
- I: is the current .

To study the effect delivered by the solar cell , we also use the formula for **electrical power** :

$$P = IV = \frac{V^2}{R} \tag{2}$$

where :

- V: Is the voltage .
- R: is the resistance .
- I: is the current .
- P: is the effect delivered by the solar cell.

3 Experimentally

In the following series of experiments ,we will use these equipments :

- A light projector.
- A voltage source.
- Two multimeter.
- A selection of resistors.
- Two identical solar cells.
- connecting wires.

3.1 Uncertainty Analysis

Instruments uncertainties:(2)

- Voltage (V): measured using a digital multimeter with an uncertainty of $\pm 0.025\%$ of the reading in V.
- Current (I): measured using a digital Ammeter with an uncertainty of $\pm 0.05\%$ of the reading .
- Resistance (R): assumed uncertainty of $\pm 0.05\%$ of the reading in (Ω) for the variable resistor

Uncertainty in power calculation :
The power P is calculated using the formula:

$$P = I \cdot V$$

To calculate the uncertainty in power ΔP, we use the standard formula of uncertainty for multiplication (1):

$$\frac{\Delta P}{P} = \frac{\Delta I}{I} + \frac{\Delta V}{V} \Rightarrow \quad \Delta P = P\left(\frac{\Delta I}{I} + \frac{\Delta V}{V}\right) \tag{3}$$

3.2 Solar cell as a semiconductor

The purpose of this experiment is to examine the difference in the behavior of a solar cell when exposed to light compared to when it is in darkness.

We begin by setting up the experiment using a one solar cell. A multimeter is connected in parallel to measure the voltage, an ammeter is connected in series to measure the current. A resistor is included in the setup, and the whole circuit is connected to a power supply, as shown in figure 2:

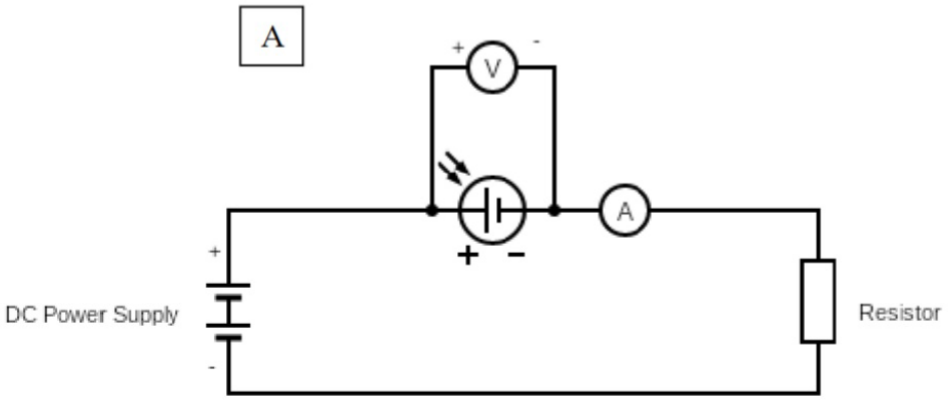


Figure 2: Setup for measuring solar cell current-voltage characteristics

The experiment will be repeated in two different situations:

- Part A: When the solar cell is exposed to light .
- Part B: :When the solar cell is in darkness.

The ammeter is set to a value of 300mA and a resistance of 10Ω.

First, we performed measurements in the absence of light,where we recorded the current I and voltage V over a voltage ranging -5V to$+5$V in increments of 0.25V.

Next, we repeated the experiment under illumination by placing the solar cell in front of a desk lamp located at at distance 50cm as shown in figure 3:

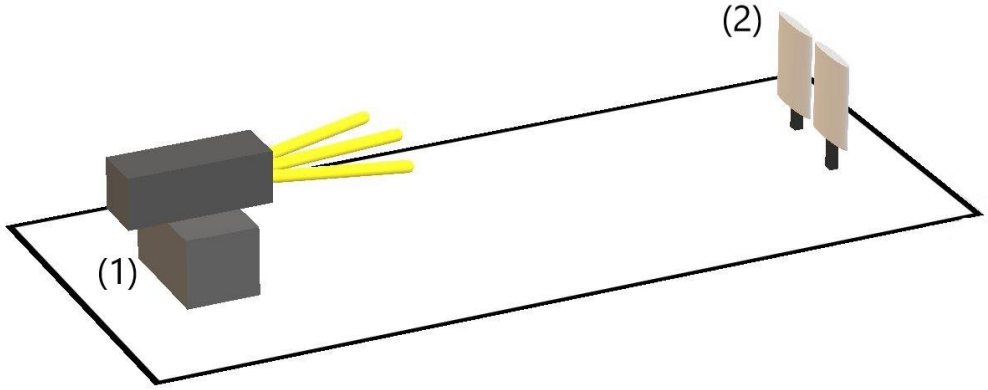


Figure 3: A solar cell illuminated by a light source

Current and voltage were measured over the same voltage range. Once all the measurements were obtained, we plotted them against each other to obtain the characteristic curve and compare it to the theoretical reference curve shown in figure 4 below:

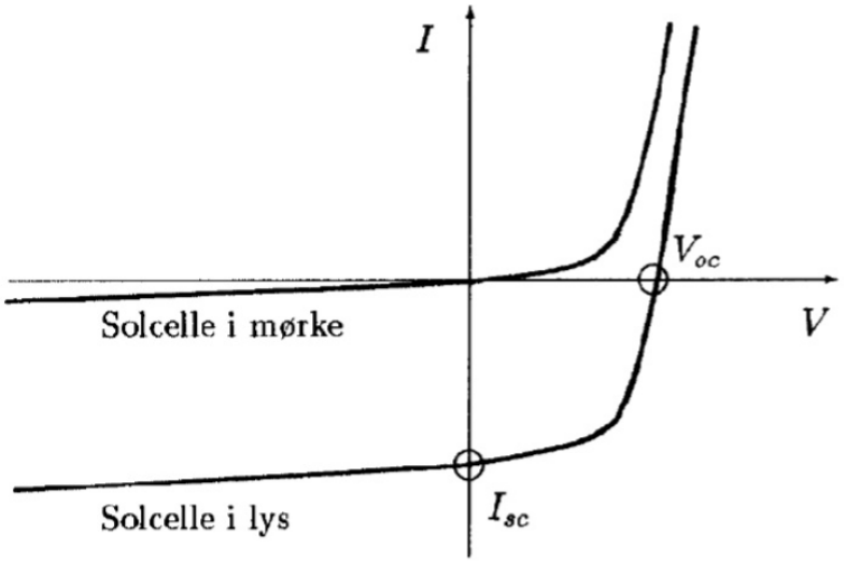


Figure 4: The theoretical IV curve for a solar cell in light vs in darkness

3.3 The Solar Cell's Optimal Load

This experiment consists of two solar cells placed in front of a projector lamp. In the first part, only one cell will be used. The cell should be placed in front of the lamp in such a way that a second cell can be added later without moving the first one.

In **Part A**, we will measure the current and voltage produced by the solar cell, without using an external voltage source or resistance.

In **Part B**, a variable resistor will be added to the circuit. The resistance will range from 0.2Ωto 10Ω.

The aim here is to use as few connecting cables as possible to minimize uncertainty in **R**

Finally, the effect can be computed using the formula (2). We then use our measurements to plot the electrical power **P** as a function of the resistance **R** in the circuit to find the value of R when the cell was the most efficient .

3.4 Combination of single solar cells into a solar panel

In this assignment, we examine solar cells connected in series and parallel configurations. Both types of configurations are often used in solar panel systems, and our goal is to investigate the advantages and disadvantages of each.

We will analyze each configuration separately as shown in the figures 5 and 6 below :

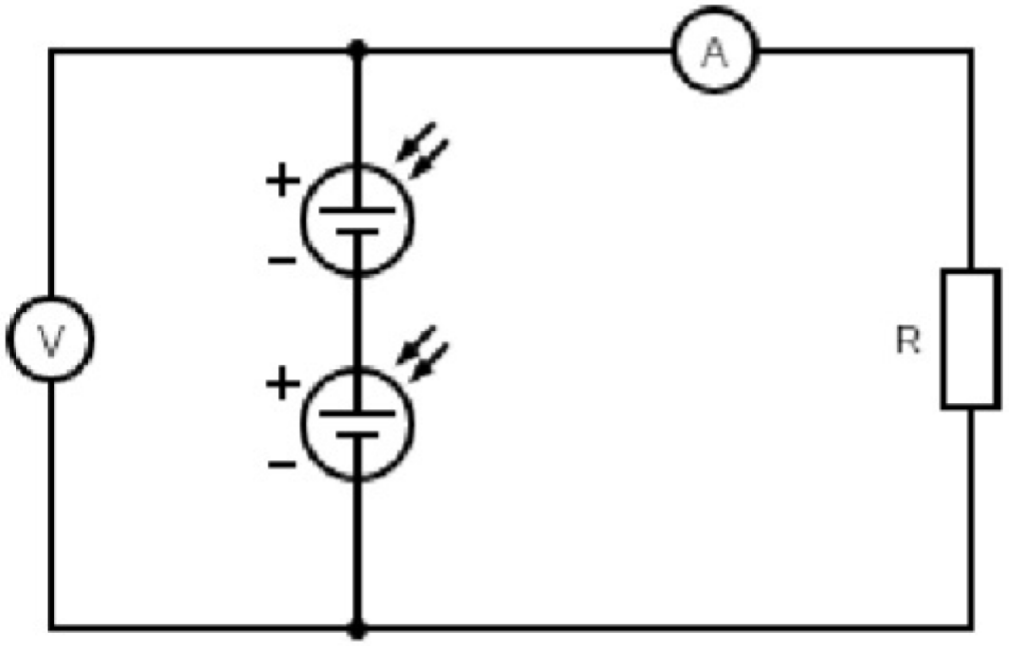


Figure 5: Two identical solar cells connected in series

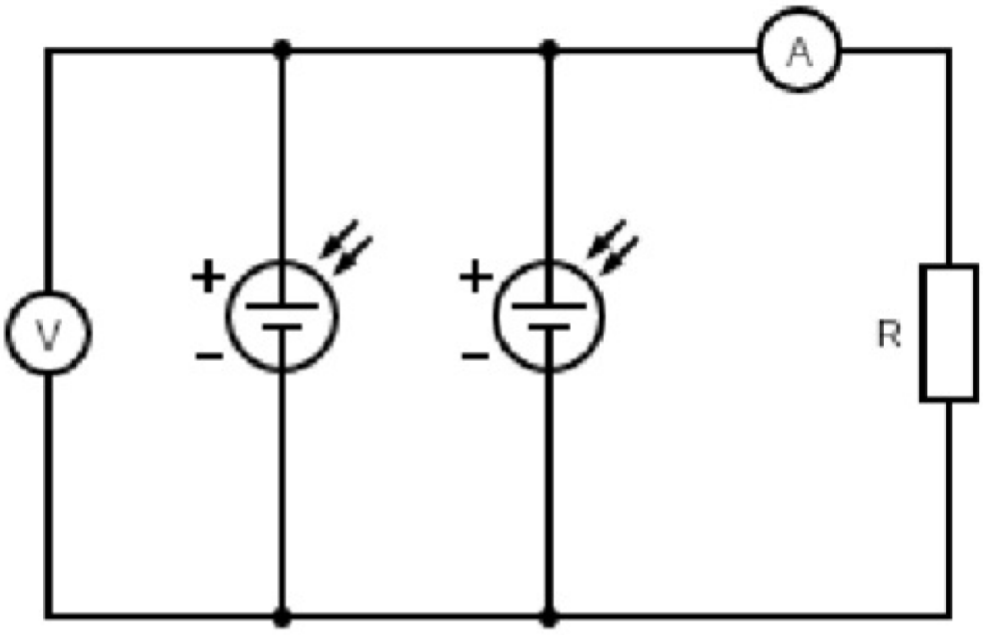


Figure 6: Two identical solar cells connected in parallel

First, we will be take measurements for both setups under full illumination. Our goal is to determine the maximum power can be produced in each case.

In part A ,we will measure the behavior of solar cells connected in series. The cells will be placed at equal distances from the light source, and we will record current and voltage measurements for different values of resistances.Then,we will plot effect **P** as a function of resistance **R**.

In part B , we will repeat the same process using parallel-connected solar cells.

3.5 Non-Uniformly illumination

In this experiment, we will examine series-connected and parallel-connected solar cells, with a difference being that we will change the illumination condition .

In real-world scenario such as installing multiple solar cells on the roof of a house or cabin ,the lighting often not even . Roofs may be partially shaded

by trees, clouds etc. In this experiment, we will study the effect of partial shading on power production and optimal load for different connection types.

To visualize this case, we will place one of the solar cells in shadow by placing an object between it and the light source, while keeping the other cell illuminated. This setup will be repeated for both series-connected and parallel-connected solar cells.

Again, we will measure the current and voltage for different **R** values, calculate the power P, and plot it as a function of resistance **R** for both cases and compare which connection type generates a better effect under different illumination conditions

Again, we will plot the power output **P** as function of resistance **R** for both cases and compare which connection type generates the better effect in none-uniformly lightning.

4 Results

4.1 Solar cell as a semiconductor

Part A: When the solar cell is exposed to light: We performed the experiment and obtained the results shown in the table 1:

$\mathbf{I_{light}}$ $\pm 0.05\%$ **(mA)**	$\mathbf{V_{light}}$ $\pm 0.025\%$**(V)**
126.4	0.52
93.9	0.51
74.6	0.50
58.8	0.49
26.63	0.47
-1.72	0.44
-25.34	0.40
-35.0	0.37
-39.4	0.35
-43.5	0.30
-47.7	0.24
-49.1	0.21
-50.5	0.15
-51.1	0.11
-51.6	0.05
-52.5	-0.49
-52.7	-1.00
-53.8	-1.93
-55.8	-3.04
-58.1	-3.94

Table 1: Measured current and voltage values for a single solar cell under illumination.

Part B: :When the solar cell is in darkness:

We performed the experiment and obtained the results shown in the table 2:

$\mathbf{I_{dark}}\pm0.05\%$ **(mA)**	$\mathbf{V_{dark}}\pm0.025\%$ **(V)**
97.8	0.503
61.4	0.475
43.8	0.453
27.66	0.421
23.90	0.411
13.08	0.361
12.46	0.357
8.00	0.319
6.27	0.298
4.85	0.277
3.72	0.257
2.45	0.227
1.56	0.199
0.92	0.170
-0.24	0.000
-0.21	-0.25
-0.35	-0.42
-0.66	-0.77
-1.01	-1.00
-1.20	-1.26
-1.47	-1.49
-1.84	-1.77
-2.17	-2.00
-2.52	-2.25
-3.01	-2.55
-3.28	-2.71
-3.78	-2.98
-4.56	-3.34
-5.32	-3.54
-5.77	-3.77
-6.36	-3.97
-7.21	-4.27
-7.76	-4.62
-8.78	-4.75
-9.57	-4.98

Table 2: Measured current and voltage values for a single solar cell in darkness.

Plugging the values from both tables into a Python program and plotting the results against each others , we obtained the following IV curve shown in the figure 7:

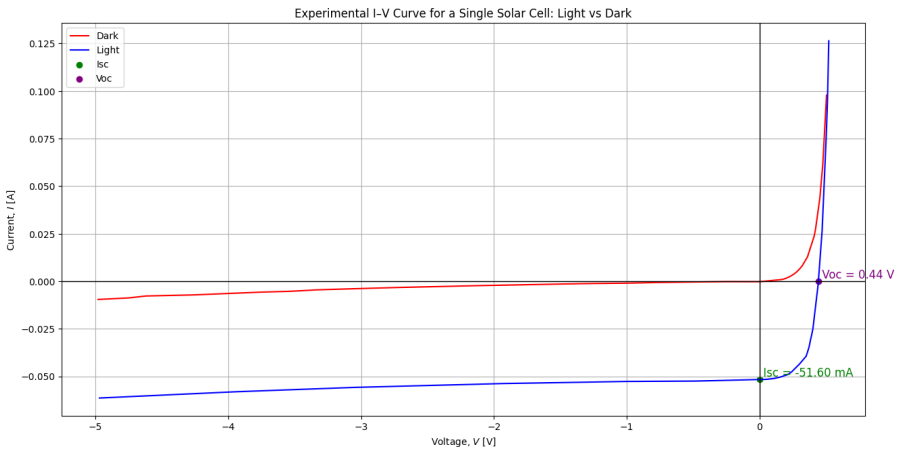


Figure 7: The experimental IV curve for a single solar cell in light vs in darkness , also showing the intersecting points

The intersecting points with I and V were calculated to be : $V_{oc} = 0.44$V , $I_{sc} = 51.6$mA.

4.2 The Solar Cell's Optimal Load

Part A : measuring the current and voltage produced by the solar cell, without using an external voltage source or resistance.
The resistance in Ammeter was $0.2 - 0.3\Omega$.

First , we measure the highest value for I and V individually without resistance:

- $I_{max} = 0.14 \pm 0.007$A.
- $V_{max} = 0.496 \pm 0.01$V.

Next , we measure the highest value for I and V simultaneously to be :

- $I_{max} = 0.141 \pm 0.007$A.
- $V_{max} = 0.009 \pm 0.01$V.

Part B: measuring the voltage and the current produced by solar cells with a variable resistance: The experiment was performed using different values of resistance R, and recorded the voltage and current measurements . The results are shown in the table 3 below:

I $\pm 0.05\%$**(mA)**	**V** $\pm 0.025\%$**(V)**	**R** $\pm 0.05\%(\Omega)$
0.14	0.015	0.0
0.139	0.043	0.2
0.139	0.071	0.4
0.14	0.099	0.6
0.139	0.126	0.8
0.139	0.149	1.0
0.138	0.176	1.2
0.137	0.203	1.4
0.137	0.215	1.5
0.136	0.229	1.6
0.136	0.240	1.7
0.135	0.253	1.8
0.134	0.265	1.9
0.133	0.276	2.0
0.128	0.329	2.5
0.12	0.373	3.0
0.119	0.376	3.1
0.117	0.384	3.2
0.115	0.389	3.3
0.113	0.394	3.4
0.111	0.400	3.5
0.103	0.417	4.0

Table 3: Measured current, voltage, and resistance values for a solar cell with variable resistance.

The corresponding P-R curve was obtained by inserting the values in the table 3 into a Python code as shown in figure 8:

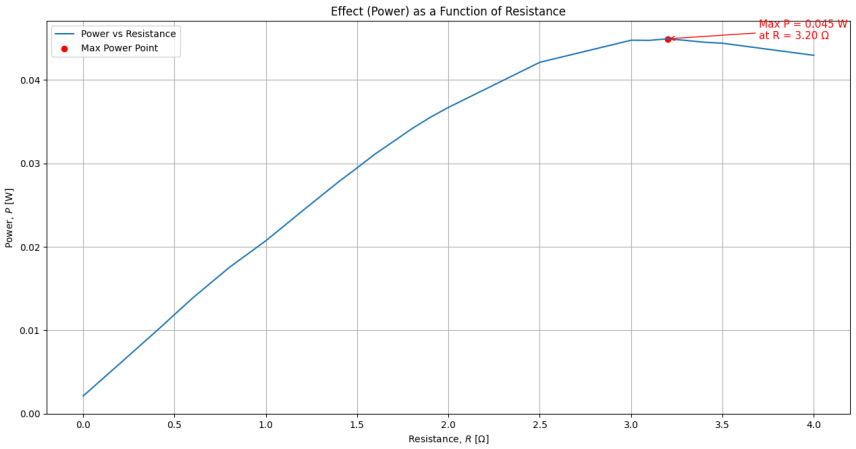


Figure 8: The P-R curve showing the effect as a function of resistance

As shown in the plot , the maximum value of P : $P_{max} = 0.043 \pm 0.0055$ W corresponds to the resistance $R = 3.2 \pm 0.16\Omega$.

4.3 Combination of single solar cells into a solar panel

Part A : solar cells connected in series. The measurements obtained for this configuration are shown in the table 4

I±0.05% **(mA)**	**V**±0.025% **(V)**	**R**±0.05% (Ω)
0.123	0.014	0.0
0.122	0.131	1.0
0.123	0.256	2.0
0.124	0.317	2.5
0.123	0.378	3.0
0.123	0.504	4.0
0.120	0.610	5.0
0.117	0.709	6.0
0.113	0.745	6.5
0.112	0.752	6.6
0.112	0.758	6.7
0.111	0.765	6.8
0.110	0.771	6.9
0.110	0.779	7.0
0.109	0.784	7.1
0.108	0.789	7.2
0.107	0.793	7.3
0.107	0.799	7.4
0.106	0.805	7.5
0.102	0.827	8.0

Table 4: Measured current, voltage, and resistance values for a solar cell connected in series .

Plotting these values to obtain the P-R curve for series connected solar cells as shown in figure 9:

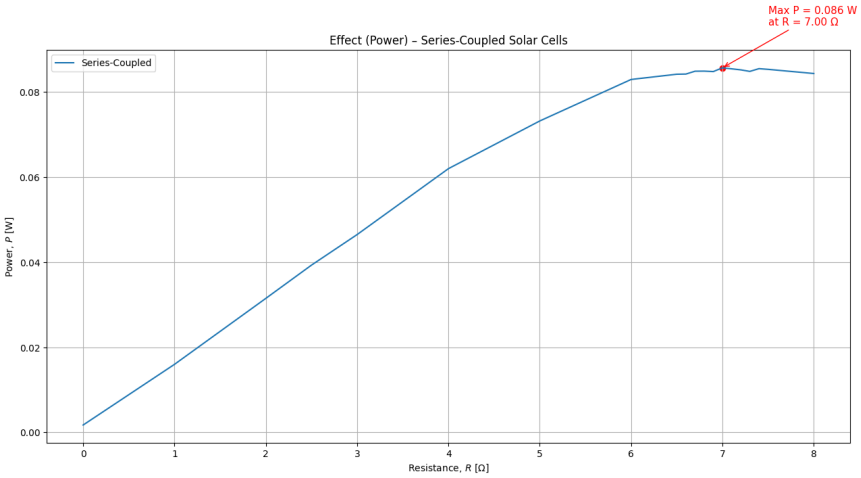


Figure 9: Plot of the effect as a function of resistance for solar cells connected in series

The maximum output of the power P is 0.086 ± 0.009W corresponds to resistance $R = 7 \pm 0.35\Omega$.

Part B : solar cells connected in parallel. The measurements obtained for this configuration are shown in the table 5

I$\pm$0.05% **(mA)**	**V**$\pm$0.025% **(V)**	**R** $\pm$0.05% (Ω)
0.264	0.038	0.0
0.252	0.278	1.0
0.246	0.319	1.2
0.232	0.370	1.5
0.227	0.384	1.6
0.220	0.395	1.7
0.215	0.405	1.8
0.208	0.414	1.9
0.200	0.421	2.0
0.171	0.445	2.5
0.148	0.457	3.0

Table 5: Measured current, voltage, and resistance values for solar cell connected in parallel .

Plotting these values to obtain the P-R curve for parallel connected solar cells as shown in figure 10:

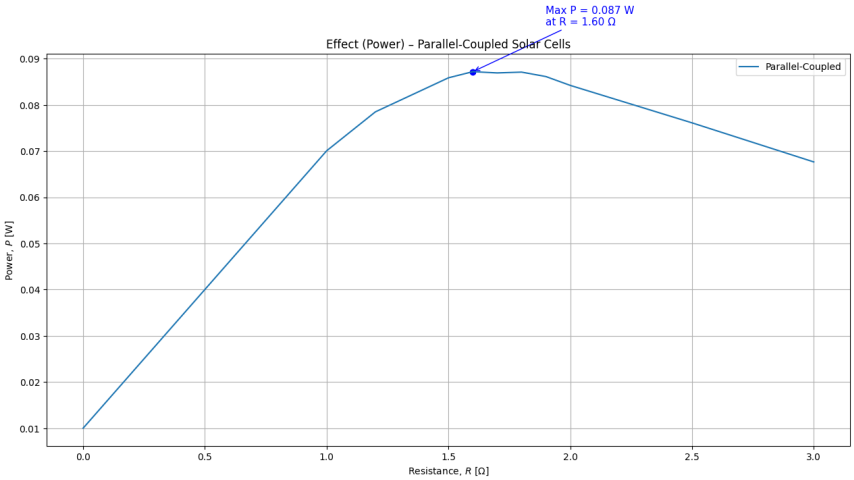


Figure 10: Plot of the effect as a function of resistance for solar cells connected in parallel.

The maximum output of the power P is 0.087 ± 0.009W corresponds to resistance $R = 1.6 \pm 0.08\Omega$.

A good way to compare the curves of parallel-connected and series-connected solar cells is to plot both curves in the same graph as shown in figure 11:

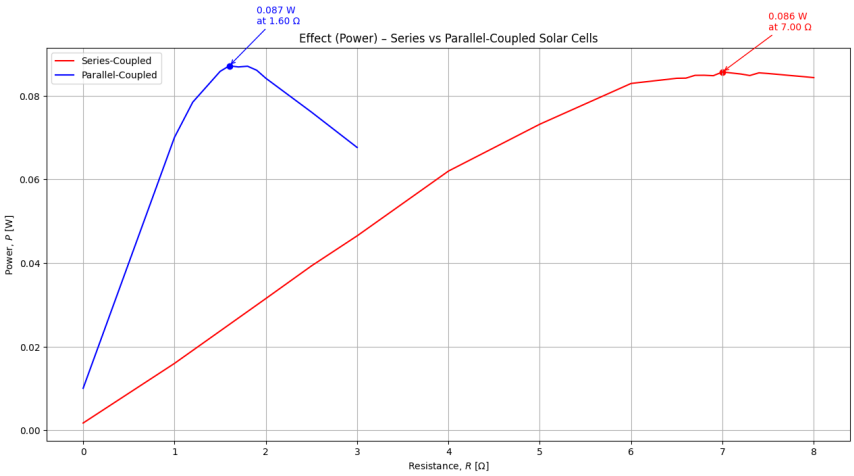


Figure 11: P-R curve for parallel-connected compared to series-connected solar cells

4.4 Non-Uniformly illumination

Part A : In this part, we measure the current **I** and voltage **V** for different values of resistance **R** of series-connected solar cells while simulating shading by partially covering one of the solar cells. The results are presented in the table 6 below

I$\pm 0.05\%$ **(mA)**	**V** ± 0.025**(V)**	**R** $\pm 0.05\%(\Omega)$
0.036	0.040	0.0
0.036	0.038	1.0
0.036	0.074	2.0
0.036	0.110	3.0
0.036	0.145	4.0
0.036	0.181	5.0
0.036	0.217	6.0
0.035	0.252	7.0
0.035	0.287	8.0
0.035	0.322	9.0
0.035	0.357	10.0
0.035	0.532	15.0
0.033	0.677	20.0
0.033	0.700	21.0
0.032	0.720	22.0
0.031	0.736	23.0
0.031	0.752	24.0
0.030	0.760	24.5
0.030	0.767	25.0
0.030	0.772	25.5
0.030	0.778	26.0
0.029	0.790	27.0
0.027	0.816	30.0

Table 6: Measured current, voltage, and resistance values for series-connected solar cells under partial shading.

Plotting these values to obtain the P-R curve for series- connected solar cells in simulated shading as shown in figure 12:

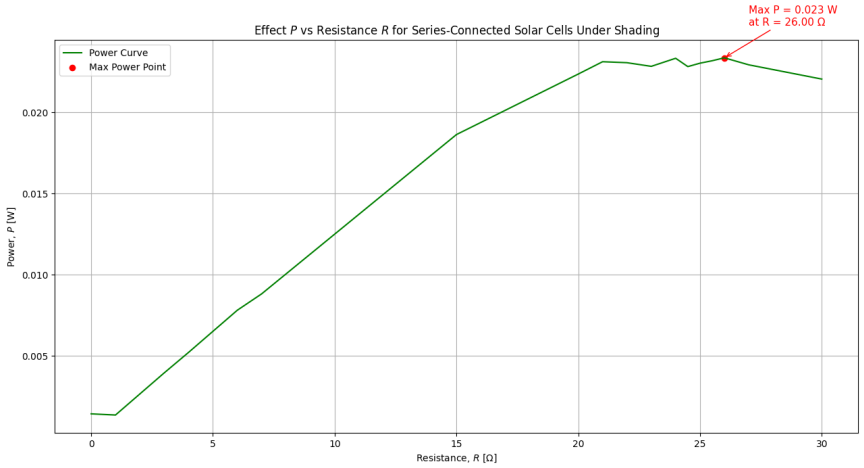


Figure 12: P-R curve for series-connected solar cells in partially shading

The maximum output of the power P is 0.023 ± 0.003W corresponds to resistance $R = 26 \pm 1.3\Omega$.

Part B : In this part, we measure the current **I** and voltage **V** for different values of resistance **R** of parallel-connected solar cells while simulating shading by partially covering one of the solar cells. The results are presented in the table 7 below:

I±0.05% **(mA)**	**V**±0.025% **(V)**	**R**±0.05% (Ω)
0.176	0.022	0.0
0.173	0.187	1.0
0.167	0.265	1.5
0.157	0.327	2.0
0.156	0.338	2.1
0.153	0.349	2.2
0.150	0.358	2.3
0.147	0.366	2.4
0.145	0.373	2.5
0.142	0.381	2.6
0.139	0.387	2.7
0.136	0.392	2.8
0.133	0.396	2.9
0.129	0.400	3.0
0.117	0.419	3.5
0.106	0.431	4.0

Table 7: Measured current, voltage, and resistance values for parallel-connected solar cells under partial shading.

Plotting these values to obtain the P-R curve for parallel- connected solar cells in simulated shading as shown in figure 13:

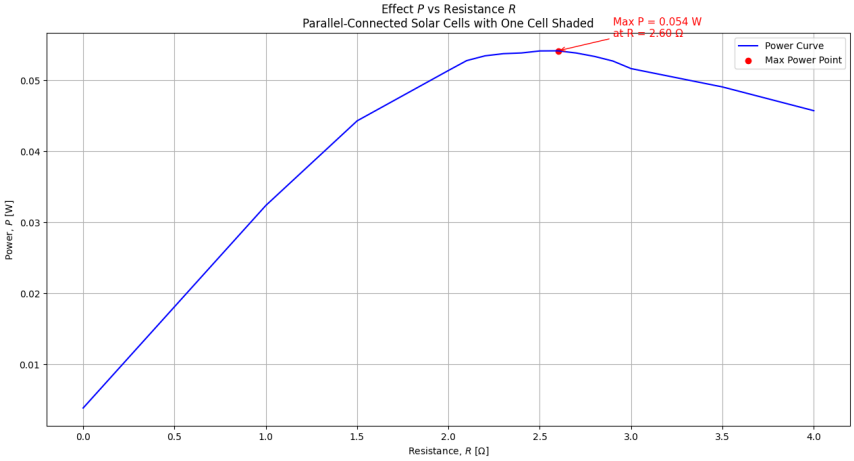


Figure 13: P-R curve for parallel-connected solar cells in partially shading

The maximum output of the power P is 0.054 ± 0.005W corresponds to resistance $R = 2.6 \pm 0.13\Omega$.

A good way to compare the curves of parallel-connected and series-connected solar cells in partial shading is to plot both curves in the same graph as shown in figure 14:

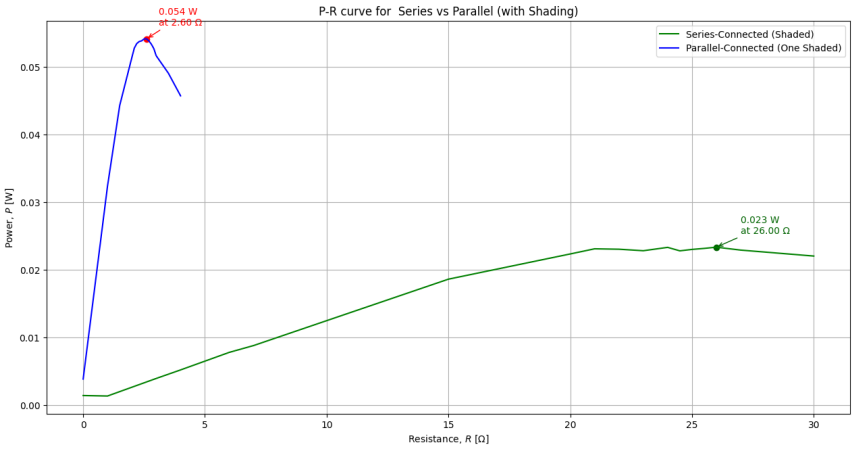


Figure 14: P-R curve for parallel-connected compared to series-connected solar cells in partial shading

4.5 Discussion

4.5.1 The solar cell as a semiconductor diode

In this part of the experiment, we measured the current and voltage produced by a single solar cell for both illuminated and dark conditions. We then plotted the experimentally obtained I–V characteristics to compare with the reference theoretical curve provided in figure4

The resulting graph, shown in Figure 7 is close to the reference theoretical curve .

Near the short-circuit current (Isc)**:** In this region,the current is mainly determined by the photocurrent generated due to the absorption of light.The current is almost constant, leading to an almost flat curve.

Near the open-circuit voltage (V_{oc})**:** In this region, the output current is close to zero, and the voltage approaches the open-circuit voltage. As the voltage changes from negative to positive, there is an exponential increase in the current with voltage. This leads to a sharp increase in the curve we see.

4.5.2 The solar cell's optimal load

In this experiment, the resulting graph is shown in Figure 8. From the graph, we can observe that the power output of the solar cell depends on the resistance of the connected load. The power increases with resistance, reaching a maximum value of approximately $P \approx 0.043 \pm 0.0055W$ at around $R \approx 3.2 \pm 0.16\Omega$, which can be considered the optimal load. After this point, the power output begins to decline.

4.5.3 Combination of single solar cells into a solar panel

In this experiment, we examined the power output of parallel-connected versus series-connected solar cells. From Figure 11, where both curves are plotted on the same graph, we observe the following: the parallel-connected solar cells achieve a maximum power effect of $P = 0.087 \pm 0.009$W when the load is $R = 1.6 \pm 0.08\Omega$,while the series-connected solar cells reach a maximum effect of $P = 0.086 \pm 0.009$W when the load is $R = 7 \pm 0.35\Omega$. By comparing the two configurations, we note that parallel-connected solar cells require lower resistance to reach their maximum power output than series-connected ones .

In comparison to the single solar cell results shown in Figure 8, we find that the series configuration produces approximately twice the power of a single cell and does so at double the optimal load $R = 7\Omega$ compared to $R = 3.2\Omega$ for single cell.

At the same time, the parallel-connected solar cells also produce about twice the power of a single cell, but at half the optimal load $R = 1.6\Omega$ compared to $R = 3.2\Omega$ for single cell.

This makes the two-solar-cell configurations, more precisely: parallel-connected solar cells , a better option than a single cell in optimal illumination conditions

4.5.4 Non-Uniformly illumination

In this experiment, we examined the power output of parallel-connected versus series-connected solar cells under non-uniform illumination conditions.

The results are presented in Figure 14, where both curves are plotted on the same graph. From this graph, we observe that in shading conditions, parallel-connected solar cells produced a maximum power of $P = 0.54 \pm 0.005$W at an optimal load of $R = 2.6 \pm 0.13\,\Omega$, while series-connected solar

cells produced a maximum power of $P = 0.23 \pm 0.003$W at an optimal load of $R = 26 \pm 1.3\,\Omega$.

This makes the parallel-connected configuration much more efficient than the series-connected configuration under non-uniform illumination. In fact, it is more effective to use a single solar cell than to connect two cells in series under shading condition .

4.5.5 Summery

	Series	**Parallel**
Uniform illumination	$P = 0.086 \pm 0.009$ W	$P = 0.087 \pm 0.009$ W
Partial shade	$P = 0.23 \pm 0.005$W	$P = 0.54 \pm 0.003$W

Table 8: Comparison of Series and Parallel under Different Conditions

4.6 Conclusion

In this set of experiments, we investigated the electrical behavior of solar cells under various conditions. Starting with a single solar cell, we measured current and voltage under both illuminated and dark conditions and in acceptable way reproduced the characteristic I–V curve, which aligned well with theoretical reference graph .

We then explored how the power output of a solar cell varies with load resistance. The results showed that power increases with resistance up to an optimal value, beyond which it declines. For a single solar cell, the maximum power achieved at approximately $R = 3.2\,\Omega$, producing around 0.043 W.

Next, we compared two solar cells connected in series and in parallel. Under uniform illumination, both configurations produced nearly twice the power of a single cell, with the series connection requiring approximately $R = 7\,\Omega$ and the parallel connection around $R = 1.6\,\Omega$ for optimal power output. These results confirm that either configuration can improve performance compared to a single cell.

Finally,we examined the behavior under non-uniform illumination (simulating partial shading). The parallel-connected solar cells produced a significantly higher maximum power output (0.54 W at $R = 2.6\,\Omega$) compared to the series-connected configuration (0.23 W at $R = 26\,\Omega$). This shows the disadvantages of series-connected solar cells in shading conditions and highlights the advantages of the parallel configuration in such conditions . In

fact, under partial shading, even a single solar cell can be a better option than two solar cells connected in series.

To summarize : based of the results presented earlier , we conclude that the parallel-connected solar cells is the optimal option regardless of the illumination conditions .

References

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